

Karl Stencell

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Professional Experience

Senior Software Engineer / Arctic Wolf

August 2022 – Present

- **Lead Architect:** Designed and built the software control layer for a multi-rack hardware lab, integrating PXE builders, VLAN-based test isolation, Jenkins automation, and supported customer device variants into a repeatable pre-production release-validation platform.
- **Validation Framework:** Designed and built a Python/pytest hardware-validation framework using composable Step, Flow, and Suite layers to express bake, manifest-upgrade, OS-upgrade, and snapshot-restore workflows across the physical hardware fleet.
- **Production Release Pipeline:** Built and operate a Go/Kubernetes release service deployed across five AWS regions, orchestrating rolling updates to 50,000+ customer devices with automatic wave ramping, 5% failure circuit breakers, and Slack alerting.
- **Tech Lead:** Drive quarterly planning, story breakdown, capacity allocation, and onboarding for the team; ramped two engineers transferring from adjacent teams through bi-weekly knowledge-transfer sessions and hit or exceeded sprint commitments across the last three quarters.
- **Developer Enablement:** Built helper scripts, operational docs, and debugging tools that let engineers query device metadata, inspect sensor status, update PXE boot files, and triage lab failures without running full Jenkins pipelines.

Co-Founder and Technical Lead / BKG Software

February 2023 – Present

- **Full-Stack Delivery:** Own architecture, delivery, deployment, and infrastructure decisions for client projects, including a Laravel/MySQL operations platform for wholesale inventory, customer orders, nightly till reconciliation, and automated S3 backups.

Projects

Upvote Approved

Laravel, Vue, FastAPI, ChatGPT API

A production product-discovery site that turns Reddit sentiment into curated buying guides: a Laravel/Vue frontend and Filament admin portal coordinate with a Python/FastAPI pipeline that searches Reddit threads, scrapes comments, scores testimonial quality with the ChatGPT API, and publishes approved products through a staging-to-production Laravel Forge deployment flow.

- **Pipeline Architecture:** Designed the data model and admin workflow for in-flight requests, partial failures, idempotent re-runs, manual review, and approval-to-publish control.
- **Production Workflow:** Built production-parity staging on Digital Ocean with separate Laravel and FastAPI instances, enabling live admin testing behind Forge access controls before promoting changes to production.
- **Traction:** Reached 600+ visits across 40 catalogued products in the first week of a Google Ads test.

Strategy Forge

Python, SQLite, Pandas, NumPy

A local quant research workbench for discovering robust trading strategies across historical OHLCV data — users define declarative JSON strategy specs, sweep parameter spaces with pluggable search algorithms, backtest candidates day-by-day with next-bar fills and commission modeling, and persist results to SQLite.

- **Architecture:** Separated strategy definition, search-space generation, simulation, and result persistence so generator algorithms can produce candidates independently of the backtesting engine.
- **Simulation Ledger:** Built a SQLite-backed experiment ledger that records each strategy specification before execution and merges results post-run, creating a queryable history of every candidate tested across parameter sweeps.

Skills

Python / PHP / Laravel / Typescript / Go / pytest / Vue / FastAPI / Docker / Kubernetes / AWS / Terraform / Ansible / Proxmox / Jenkins / Linux / MySQL / SQLite / Neo4j / Grafana

Education / Certifications

Bachelor of Computer Science

Conestoga College, Ontario

Bachelor of Science (Honours) Theoretical Physics

University of Guelph, Ontario

AWS Certified Cloud Practitioner

2024